Product rule

1

a $u' = 6x^5$

b $v' = 4x^3$

c $\dfrac{dy}{dx} = 10x^9 + 24x^5$

2

a $\dfrac{du}{dx} = 7x^6$

b $\dfrac{dv}{dx} = 6x^5$

c $\dfrac{dy}{dx} = 13x^{12} + 30x^5 + 42x^6$

3

a $\dfrac{dy}{dx} = 5x^4 + 27x^2$

b $\dfrac{dy}{dx} = 5x^4 + 27x^2$

4

a $f'(x) = 24x - 23$

b $g'(t) = -8t^3 + 18t^2 + 3$

c $g'(y) = 42y^5 - 144y^3 + 10y$

d $f'(x) = 14x^{\frac{4}{5}} + 4x^{\frac{1}{3}} + 54x^{\frac{1}{2}} + 9x^{-\frac{1}{2}}$

e $f'(x) = -\dfrac{1}{2}x(3-x)^{-\frac{1}{2}} + (3-x)^{\frac{1}{2}}$

f $f'(x) = \dfrac{10}{3}x^{\frac{2}{3}} - \dfrac{8}{3}x^{\frac{5}{3}}$

g $f'(x) = -\dfrac{2}{3}x^{\frac{1}{3}}(1-x)^{-\frac{1}{3}} + \dfrac{1}{3}(1-x)^{\frac{2}{3}}x^{-\frac{2}{3}}$

h $\dfrac{dy}{dx} = -4x - \dfrac{5}{2}x^{\frac{3}{2}} + 3x^{-\frac{1}{2}}$

i $\dfrac{dy}{dx} = -2x - \dfrac{3}{x^2}$

j $\dfrac{dy}{dx} = x^2(5x+3)^6(50x+9)$

k $\dfrac{dy}{dx} = 6x^4\left(x^2+3\right)^2\left(11x^2+15\right)$

l $\dfrac{dy}{dx} = 3\left(x^2+x+1\right)^8\left(19x^2+10x+1\right)$

m $\dfrac{dy}{dx} = 5(5x+7)^6(8x-9)^4(96x-7)$

n $\dfrac{dy}{dx} = \dfrac{18x+19}{\sqrt{5+4x}}$

o $\dfrac{dy}{dx} = 8(5+8x)^{\frac{3}{4}}(22x+5)$

p $\dfrac{dy}{dx} = \dfrac{32x^5}{\sqrt{8x+3}} + 40x^4\sqrt{8x+3}$

q $\dfrac{dy}{dx} = \dfrac{3(2x+3)}{\sqrt{x+1}}$

r $\dfrac{dy}{dx} = \dfrac{4(3x-1)}{\sqrt{1-2x}}$

5

a $y' = 40x - 23$

b $f'(x) = 100x^4 - 92x^3 + 18x^2$

6

a

i $u = x, v = (x-8)^4$

ii $\dfrac{dy}{dx} = (x-8)^3(5x-8)$

iii $x = \dfrac{8}{5}, 8$

b

i $u = x^3, v = (x+3)^4$

ii $\dfrac{dy}{dx} = x^2(x+3)^3(7x+9)$

iii $x = -3, -\dfrac{9}{7}, 0$

c

i $u = x+2, v = (x+5)^6$

ii $\dfrac{dy}{dx} = (x+5)^5(7x+17)$

iii $x = -\dfrac{17}{7}, -5$

7 $n = 4$

Gradients, tangents and normals

8

a 0

b 15

c 135

9 $g'(3) = -54$

10

a $f'(x) = 2(x+3)^2(2x+3)$

b $y = 8(x+1)$

c $y = -\dfrac{1}{8}(x+1)$

11 $x = -2 \pm \dfrac{\sqrt{30}}{3}$